1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 analysis highlights that high-risk NMIBC cohorts have upregulated LAG3 yet organized local immune structures (TLS/TILs) often predict better immune control in other cancers; why then might TLS-like lymphoid aggregates not translate into effective anti-tumor immunity or therapy response in high-risk NMIBC? 2) Tier 1 paths used: (a) NMIBC → high-risk group → high LAG3 expression (direct mechanistic backbone linking risk to immune-checkpoint–mediated exhaustion); (b) the PSA/finasteride → PSA density axis was reviewed but deliberately deprioritized because it addresses a separate screening problem and would dilute the mechanistic focus. 3) Tier 2 evidence supporting the claim: urothelial carcinoma as the histologic substrate grounds NMIBC biology; the differential expression of LAG3 across cancers (e.g., low in READ) provides tissue-specific context suggesting NMIBC-specific immune checkpoint dominance; generated-image/AI paths imply that H&E–derived TLS/TIL imaging features are detectable and can be used to spatially map immune organization. 4) Why therapy_response as the endpoint: the mechanistic question concerns functional immune competence and predictiveness of local immune architecture for benefit from immunotherapy (i.e., does TLS presence actually predict response?), so a therapy-response proxy is the most direct and actionable clinical endpoint. 5) What I excluded and why: I excluded the PSA/finasteride/PSA-density core path from the primary hypothesis to keep a single, sharply focused mechanistic question (immune evasion in NMIBC). Other exploratory items (PSA density, finasteride, READ) are listed as secondary_modifiers for future cross-cancer comparisons but are not part of the central hypothesis?

Working Hypothesis

Primary hypothesis: In high-risk NMIBC, TLS-like lymphoid aggregates exist but functionally fail to support effective anti-tumor immunity when LAG3 expression is high, due to T cell exhaustion within TLS niches. Consequence: TLS-rich NMIBC tumors with high LAG3 will demonstrate reduced response to standard intravesical immunotherapy compared with TLS-low or low-LAG3 NMIBC. Secondary hypothesis: Therapeutic strategies that block LAG-3 or otherwise alleviate exhaustion within TLS will restore cytotoxic T cell activity in the TLS microenvironment and improve therapy response. Testable predictions: (a) Within TLS, high LAG3+ T cells co-localize with exhaustion markers (PD-1, TIM-3) and show reduced cytotoxic gene signatures (e.g., GZMB, PRF1) relative to TLS with low LAG3. (b) Patients with TLS-rich, high-LAG3 NMIBC exhibit lower objective response rates and shorter recurrence-free survival after intravesical immunotherapy than TLS-rich, low-LAG3 or TLS-poor NMIBC. (c) Ex vivo or in vitro assays treating NMIBC-derived T cells with LAG-3 blockade will enhance proliferation and cytotoxic function specifically in TLS-associated T cells. (d) Spatial transcriptomics will reveal distinct TLS microdomains where exhaustion programs predominate in high-LAG3 contexts, contrasting with more functional TLS zones in low-LAG3 contexts. Implications: If true, patient stratification by TLS architecture and LAG3 status could guide use of LAG-3–targeted or combination immunotherapies in high-risk NMIBC.” ,

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	LAG3, NMIBC
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

LAG3: LAG3 yet organized local immune structures (TLS/TILs) often predict better immune control in other cancers; why then might TLS-like lymphoid aggregates not translate into effective anti-tumor immunity or therapy response in high-risk NMIBC? 2) Tier 1 paths used: (a) NMIBC → high-risk group → high LAG3 expression (direct mechanistic backbone linking risk to immune-checkpoint–mediated exhaustion); (b) the PSA/finasteride → PSA density axis was reviewed but deliberately deprioritized because it addresses a separate screening problem and would dilute the mechanistic focus.

NMIBC: NMIBC cohorts have upregulated LAG3 yet organized local immune structures (TLS/TILs) often predict better immune control in other cancers; why then might TLS-like lymphoid aggregates not translate into effective anti-tumor immunity or therapy response in high-risk NMIBC? 2) Tier 1 paths used: (a) NMIBC → high-risk group → high LAG3 expression (direct mechanistic backbone linking risk to

immune-checkpoint–mediated exhaustion); (b) the PSA/finasteride → PSA density axis was reviewed but deliberately deprioritized because it addresses a separate screening problem and would dilute the mechanistic focus.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PSA density	Supporting modifier from knowledge graph
Secondary	Finasteride	Supporting modifier from knowledge graph
Secondary	Urothelial carcinoma	Supporting modifier from knowledge graph
Secondary	READ	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- PSA density: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Finasteride: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.